

Multilingual multimodal cyberbullying detection through adaptive and hierarchical fusion[☆]

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ABSTRACT

Detecting cyberbullying in multimodal content (such as memes) is challenging due to complex interactions between images and text, often involving sarcasm, multilingual usage, and other noisy real-world factors. This paper presents a multilingual multimodal cyberbullying detection framework that combines early fusion, late fusion, and hierarchical fusion strategies within a unified architecture. The framework introduces three key modules: Adaptive Cross-Modal Token Integration (ACTI) for iterative early fusion, Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG) for dynamic late fusion based on input reliability, and a Hierarchical Contextual Fusion Network (HCFN) that feeds early fused context back into later unimodal processing for refined predictions. Our system leverages state-of-the-art pretrained vision-language models (e.g., CLIP for images and XLM-RoBERTa for text) to learn subtle cross-modal representations (e.g., sarcasm or image-text irony) and uses uncertainty modeling to handle ambiguous or noisy inputs. We evaluate the approach on two benchmark datasets: the English-language Facebook Hateful Memes and the ArMeme dataset of Arabic memes. Experimental results show that our model outperforms multiple baselines (including single-modality models and a strong CLIP-based multimodal baseline), achieving high accuracy, F1-scores, and area under ROC (AUROC) across languages. Notably, it achieves state-of-the-art performance (e.g., 0.85 F1 and 0.88 AUROC on Hateful Memes), surpassing prior fusion methods. The proposed framework represents a significant step toward generalizable, culturally aware, and robust multimodal cyberbullying detection suitable for deployment across diverse social media contexts.

1. Introduction

Cyberbullying has evolved into a worldwide issue within online communities, increasingly showing through complex multimodal formats such as memes that combine visual imagery with text [1,2]. The context-dependent nature of this content poses significant challenges for automated detection systems, as offensive meanings often emerge not from the individual modalities but from the complicated, often sarcastic or ironic, interplay between them [3,4]. Recent benchmarks like the Facebook Hateful Memes challenge have demonstrated that unimodal approaches analyzing text or images separately achieve only moderate performance (F1-scores 60%–65%), fundamentally failing to capture the semantic relationships characterizing meme-based harassment [5]. This recognition has encouraged comprehensive research into multimodal fusion strategies for the detection of cyberbullying [6,7].

However, existing fusion architectures face critical limitations that fundamentally limit their effectiveness. Current approaches typically

adopt either early fusion, which combined features at the input level, or late fusion, which combines predictions from independently trained models [8]. Early fusion methods struggle to model hierarchical cross-modal relationships because merging representations before full processing loses opportunities for modality-specific processing [9,10]. Simple concatenation cannot capture the iterative learning required for understanding sarcasm, where meaning emerges through progressive acceptance of conflicting visual and textual signals [11,12]. Recent attention-based early fusion methods perform only single-pass integration, lacking iterative improvement for complex semantic conflict [13].

In contrast, late fusion maintain modality-specific processing but fails to capture essential feature-level interactions [8]. Processing visual and textual content independently until the final decision stages cannot ground visual concepts in the textual context or vice versa, missing cross-modal dependencies critical for detection [5]. While hybrid approaches attempt combining strategies, they lack principled

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mechanisms for hierarchical integration across multiple processing stages [14]. Beyond fusion strategy limitations, existing approaches assume equal reliability across modalities, applying fixed weighting regardless of input quality [15]. This assumption fails with real-world noisy inputs such as low-quality images, blurred visuals, or unreliable OCR extractions. When one modality provides misleading signals, fixed fusion cannot adaptively adjust reliance on the trustworthy modality, leading to incorrect predictions [16]. Recent work suggests that explicitly modeling uncertainty and dynamically weighting modality contributions is crucial, yet most detectors lack such mechanisms [17]. Furthermore, existing fusion architectures have not been validated on multilingual datasets or culturally diverse content. Memes with Arabic text require processing dialectal variations, right-to-left script, and culturally specific references, yet current approaches have not sufficiently addressed cross-lingual generalization [18].

To address these fundamental multimodal fusion limitations, we propose a novel hierarchical fusion framework that systematically integrates multiple fusion strategies across different processing stages [19]. Our approach is motivated by three key insights: effective cross-modal understanding requires iterative improvement rather than single-pass integration [20]; optimal fusion employs different strategies at different abstraction levels (token-level integration, hierarchical feedback, and decision-level ensemble) [21,22]; and fusion must explicitly model and adapt to input quality variations through uncertainty-aware weighting [17].

Specifically, our framework introduces three integrated innovations. First, an Adaptive Cross-Modal Token Integration (ACTI) module performs iterative early fusion through multiple rounds of cross-attention between learnable fusion tokens and visual and textual features, creating progressive bidirectional improvement necessary for detecting irony [23,24]. Unlike single-pass early fusion, ACTI enables iterative improvement across multiple cycles. Second, a Hierarchical Contextual Fusion Network (HCFN) implements recursive feedback where fused contextual representations are fed back into subsequent unimodal encoder layers, allowing globally fused context to determine local features and bridging the gap between early and late fusion [25,26]. Third, a Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG) module dynamically weights contributions from text, image, and fused branches based on learned uncertainty estimates and contextual quality signals (OCR confidence, image clarity, text length), providing robustness against noisy conditions [15].

To ensure cross-lingual applicability, we employ XLM-RoBERTa as our multilingual text encoder and CLIP’s image encoder for culturally diverse visual content [27]. We evaluate our framework on the Facebook Hateful Memes (English) and ArMeme (Arabic) datasets [28] (as shown in Table 1), achieving state-of-the-art results that validate the effectiveness and cross-lingual generalization of our hierarchical fusion architecture.

Our main contributions are:

1. We propose the first comprehensive hierarchical fusion framework that systematically integrates early (ACTI), intermediate hierarchical (HCFN), and late (CAE-UAG) fusion strategies for multimodal cyberbullying detection.
2. We introduce a novel Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG) module that explicitly models prediction uncertainty and incorporates contextual quality signals (OCR confidence, image quality scores, language indicators).
3. We design a multilingual architecture leveraging XLM-RoBERTa and CLIP that achieves strong cross-lingual performance without language-specific modifications.
4. We conduct extensive experimental validation including comprehensive ablation studies that quantify each component’s contribution.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 details the proposed framework strategies. Section 4 describes the experimental setup. Section 5 presents results and analysis to address ethical considerations, including bias mitigation. Section 6 concludes the paper.

2. Related work

Early approaches to automated cyberbullying or hate speech detection were largely text-based, leveraging techniques from natural language processing and classical machine learning. These included simple keyword spotting and lexicon-based methods, as well as training classifiers (e.g., SVMs, logistic regression) on features like n-grams, profanity dictionaries, and sentiment scores. Such methods could flag obvious insults but struggled with complex distinctions and the context-dependent nature of bullying language (e.g., sarcasm, slang, or coded phrases) [29]. The advent of deep learning led to the use of RNN/LSTM architectures and then transformer-based models (like BERT) fine-tuned on abusive language datasets, which significantly improved detection by better capturing linguistic context and variations of offensive language ([30]; [31]; [32]; [33]). Recent works have also explored detection in languages other than English: for example, systems targeting Hindi, Bengali, or other languages by training language-specific models or using multilingual embeddings have been reported to achieve high within-language accuracy [34]. However, cross-lingual and multilingual settings remain challenging due to limited labeled data in many languages and cultural differences in how bullying is expressed.

In the multimodal domain, the Facebook Hateful Memes Challenge encourage research on detecting hate speech in memes [5]. Initial baselines in that challenge showed that neither the image nor text alone was sufficient, prompting the need for multimodal fusion [35]. A variety of fusion strategies have been explored. Early fusion methods directly concatenate image and text features before classification, whereas late fusion ensembles combine separate image-only and text-only model predictions. Both strategies have limitations, early fusion can drown out modality-specific signals if not done carefully, and naive late fusion might ignore subtle cross-modal interactions. More advanced multimodal models have since been applied, many leveraging large vision-language transformers [36,37]. For example, the CLIP model [38] provides powerful joint representations of images and text and has been fine-tuned for hateful meme classification in works like Hate-CLIPper [39]. Hate-CLIPper uses CLIP’s dual encoders with an interaction module and achieved strong performance (around 85%–87% AUROC) on Hateful Memes, even surpassing human performance on some metrics. Other transformer-based multimodal approaches such as VisualBERT, ViLBERT, and UNITER integrate image and text within transformer architectures and have shown competitive results [40]. These large pre-trained models typically yield AUROC in the 0.75–0.85 range on Hateful Memes ([40]; [41]), depending on the technique. However, they often come with massive parameter counts and heavy computational requirements (e.g., some recent models count in the billions of parameters) ([41]; [42]). In practical settings, this raises concerns about efficiency and deployability.

Beyond using off-the-shelf multimodal models, researchers have proposed custom fusion techniques to better exploit cross-modal cues. Cross-modal attention mechanisms allow models to focus on relevant parts of the image given the text (or vice versa). For instance, bilinear pooling and co-attention have been used to capture fine-grained image-text interactions in hateful meme detection. More recently, there is interest in dynamic fusion strategies that adapt to input content. For example, an Uncertainty-Aware Cross-Modal Alignment (UCA) model was proposed by [43] for hate speech, which aligns image and text features while accounting for uncertainty. [44] introduced Hate UDF, an explainable hateful meme detector with an uncertainty-aware dynamic fusion approach however, their method uses threshold-based rules to decide which modality to trust. Ensemble of experts frameworks and

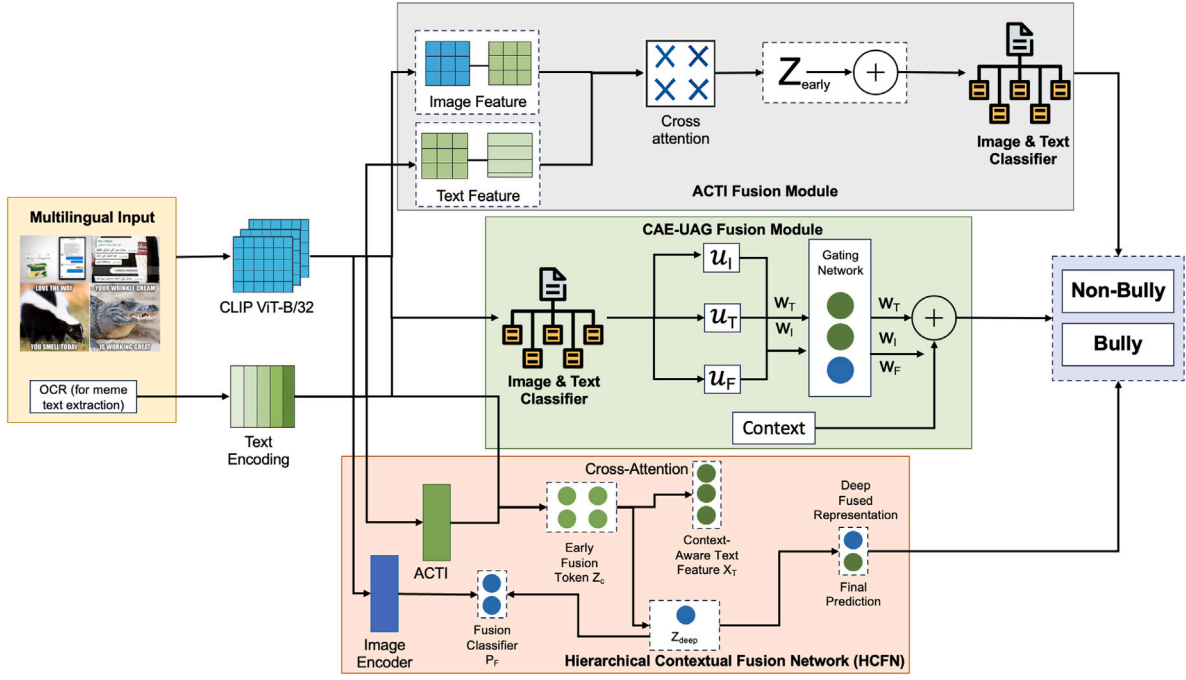


Fig. 1. The proposed platform for multilingual multimodal cyberbullying detection framework.

gating mechanisms have been explored in related contexts to weigh modalities differently per instance [43]; [44]. These works inspire our approach to incorporate uncertainty-aware gating, but our model differs by using a learned neural gating function conditioned on rich contextual signals rather than fixed thresholds.

There is a clear need to extend multimodal hate detection to languages beyond English. Prior multimodal work has been predominantly English-centric, with only a few exceptions. Bui et al. (2024) introduced a Multi3Hate dataset and approach covering multiple languages and cultures, demonstrating the difficulty of generalizing across them [45]. Kumar (2025) studied cyberbullying detection in Hinglish (code-mixed Hindi-English) text using multilingual transformers, highlighting the value of multilingual language models in detecting abuse in mixed languages [46]. Nonetheless, truly unified frameworks that handle both multilingual text and images in a combined manner are scarce. Our work addresses this by using multilingual text encoders (XLM-R) and training data to simulate code-mixed and translated scenarios, thereby training a single model to operate across languages. Additionally, cultural context in memes (e.g., region-specific slang or symbols) can be as important as language. Some recent studies attempt to incorporate external knowledge or cultural context for hate speech detection, but this remains an open challenge.

Given this background, our work stands at the intersection of these threads, we build upon lessons from prior fusion methods and uncertainty-based ensembles, leverage powerful pretrained vision-language representations (CLIP and XLM-R), and push into new territory by tackling multilingual memes with a unified, robust architecture. To the best of our knowledge, this is the first study to present a single framework that handles multilingual multimodal cyberbullying detection with an extensive integration of fusion techniques, validated on large-scale datasets in two languages. In the next section, we detail our proposed methodology and how it advances beyond the prior art.

3. The proposed platform for multimodal fusion framework

Our proposed platform is a multimodal fusion framework that processes meme images and text through multiple stages of integration. Fig. 1 provides an overview of the architecture. First, both image

and text inputs undergo preprocessing and feature extraction. The text (which may be in English, Arabic, or a code-mixed combination) is normalized and encoded via a multilingual transformer, while the image is processed by a deep convolutional or transformer-based network; we also use auxiliary tools like OCR to extract any text embedded in the image and other high-level visual context.

The core of the model consists of three fusion modules applied in sequence: (1) An early-fusion module (ACTI) that produces an initial joint representation of the text and image features; (2) A stage of hierarchical context fusion (HCFN) where the fused context is fed back into subsequent layers of the unimodal encoders (influencing their representations) and then fused again; and (3) A late-fusion ensemble module (CAE-UAG) that combines the predictions of the image branch, text branch, and fused branch by weighting them according to learned uncertainty and context signals. This hierarchical design allows the model to learn cross-modal interactions at multiple levels: from low-level feature token mixing to high-level decision ensembling. It provides both fine-grained alignment between modalities and robust aggregation of evidence needed for complex meme understanding, crucial for content across different languages and cultural contexts. In the following subsections, we describe each component in detail.

3.1. Preprocessing and feature extraction

We handle multi-language text by employing a multilingual encoder. Specifically, we use XLM-RoBERTa (XLM-R), a state-of-the-art transformer model trained on 100+ languages, to obtain text embeddings. This is critical for supporting the multilingual scope (English and Arabic) within a shared semantic space. We perform standard text preprocessing: for English, lowercasing and punctuation normalization; for Arabic, removing diacritics and stemming where appropriate, and inserting special language tokens. If a meme's text is code-mixed, we rely on XLM-R's capacity and also flag such cases with a code-mixing indicator used as a contextual feature.

Furthermore, during training, we integrate Multilingual and Culturally Adaptive techniques, including Code-Mixing Synthesis and Translation-Based Paraphrasing (back-translation), to increase linguistic robustness and coverage of slang/idioms. The output of the text encoder is an initial text feature vector $X_T \in \mathbb{R}^d$.

Table 1
Dataset summary.

Dataset	Language	Total samples	Cyberbullying samples	Non-Cyberbullying samples
ArMeme	Arabic	15 000	3544	11 456
Hateful Memes	English	10 000	3000	7000

Each meme image is first resized and normalized. We apply Optical Character Recognition (OCR) to extract any embedded text; the OCR confidence is used as a reliability signal. The image itself is encoded using the CLIP model's image encoder (ViT-B/32), chosen for its strong zero-shot representations and visual-language alignment capabilities, which aid in decoding culturally specific visual concepts. We also apply Semantic-Preserving Visual Transformations during training to make the image encoder robust to noise and varying quality common in real-world cultural data. This encoder produces a vector representation $X_I \in \mathbb{R}^d$ for the image. Additionally, we record an image quality score to later inform the model if an image is potentially not clear. After this stage, we have X_T and X_I , alongside a set of contextual metadata C : OCR confidence, image quality score, text length, and language/code-mix flags. These are incorporated in the fusion modules, particularly in the gating mechanism.

3.2. Adaptive Cross-Modal Token Integration (ACTI)

The Adaptive Cross-Modal Token Integration (ACTI) is our proposed deep early fusion module. Unlike simple concatenation of image and text features, ACTI performs an iterative token-level fusion through cross-attention, allowing fine-grained interactions between modalities from the very start. The intuition is to simulate a “conversation” between image and text representations: at each iteration, ACTI's learnable tokens attend to textual features to update their understanding, and then attend to visual features, progressively refining a joint representation that encodes subtle image–text relationships.

Specifically, we initialize a set of fusion tokens $F^{(0)} \in \mathbb{R}^{n_F \times d}$ (with n_F tokens of dimension d). We then perform L rounds of cross-modal attention. In one iteration, the updates are defined as follows:

Text-to-Fusion update: Each fusion token attends to the textual embedding sequence X_T to update the tokens:

$$F_T^{(l)} = \text{softmax}\left(\frac{F^{(l-1)} X_T^\top}{\sqrt{d}}\right) X_T + F^{(l-1)} \quad (1)$$

Image-to-Fusion update: The text-infused tokens then attend to the image features X_I :

$$F_I^{(l)} = \text{softmax}\left(\frac{F_T^{(l)} X_I^\top}{\sqrt{d}}\right) X_I + F_T^{(l)} \quad (2)$$

We repeat these steps for $l = 1$ to L (we set $L = 2$ or 3 in experiments). The final output of ACTI is an early fused representation $Z_{\text{early}} = \text{ACTI}(X_T, X_I)$ capturing the initial alignment between image and text. This deep early fusion allows the model to uncover latent correspondences (e.g., in cases of sarcasm) that unimodal encoders would miss.

3.3. Hierarchical contextual fusion network (HCFN)

The HCFN is responsible for iterative context improvement. First, Z_c , a coarse joint context vector (e.g., pooled ACTI tokens), is obtained. This Z_c is intended to capture the high-level context (e.g., the topic and likely sentiment).

Contextual Unimodal Encoding: We feed Z_c back into the text and image encoders to influence their next layer of representations via attention or gating:

$$X'_T = \text{TextEncoder}(X_T; Z_c) \quad (3)$$

$$X'_I = \text{ImageEncoder}(X_I; Z_c) \quad (4)$$

This step makes the unimodal features context-aware; for instance, the Arabic text encoder might weigh certain dialectal words differently if Z_c indicates the image context is about a sensitive topic. **Deep Fusion:** We then apply a more involved fusion network on the refined features X'_T and X'_I to get a deeper fused representation Z_{deep} . **Branch Predictions:** We produce three outputs: p_T (from X'_T), p_I (from X'_I), and p_F (from Z_{deep}), which serve as the experts for the subsequent gating module. The HCFN ensures a recursive feedback loop: early fusion informs later processing, and later fusion integrates the refined features, addressing different types of information complexity in stages.

3.4. Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG)

The role of the CAE-UAG module is to perform an intelligent late fusion of the expert predictions by weighting them according to learned uncertainty estimates u_m and the contextual features C . Formally, the gating network takes the prediction-uncertainty pairs (p_T, u_T) , (p_I, u_I) , (p_F, u_F) and the context C as input, and outputs weights w_T, w_I, w_F for each branch via a small feed-forward network. This network computes an intermediate score s_m :

$$s_m = W_c \cdot C + W_u \cdot u_m + b_m \quad (5)$$

where C incorporates signals like OCR confidence and code-mix flags, allowing the gate to dynamically adjust reliance on each modality. These scores are passed through a softmax to produce a normalized weight w_m . This structure enables the model to hedge its bets; for example, if the image classifier's uncertainty u_I is high due to detected blurriness, w_I is reduced. The final prediction probability is then:

$$P_{\text{final}} = w_T p_T + w_I p_I + w_F p_F \quad (6)$$

This uncertainty-aware ensemble is critical for robustness across diverse cultural data, relying on the modality that seems more trustworthy for a given meme.

3.5. Training objective and implementation

The model is trained end-to-end using a standard binary cross-entropy on the final output P_{final} . Additionally, we employ auxiliary losses on the intermediate outputs p_T, p_I, p_F to encourage each expert branch to learn meaningful predictions, with a total loss $L = L_{\text{final}} + \lambda(L_T + L_I + L_F)$, where $\lambda = 0.2$. We implemented the framework in PyTorch, initializing the text encoder from xlm-roberta-base and the image encoder from CLIP's ViT-B/32. We use the AdamW optimizer with differential learning rates for the pretrained encoders and fusion components, and the learning rate is scheduled with a linear warm-up followed by a cosine decay. We train for up to 30 epochs on each dataset, employing early stopping based on validation F1 score.

4. Experimental setup

4.1. Datasets and preprocessing

We evaluate our proposed framework on two benchmark datasets for multimodal cyberbullying detection across different languages and cultural contexts. We use the Facebook Hateful Memes challenge data-

Table 2
Results hateful memes.

Type	Model Variant	Validation F1-score	AUROC	Test F1-score	AUROC
Unimodal	Image-Region	52.53	57.24	52.36	57.74
	Text BERT	58.27	65.05	62.80	69.00
Multimodal (Unimodal Pretraining)	Late Fusion	59.39	65.07	63.20	69.30
	Visual BERT	65.01	74.14	66.67	74.42
Multimodal (Multimodal Pretraining)	Visual BERT COCO	65.93	74.14	69.47	75.44
No early fusion	without ACTI	76.18	83.01	78.44	84.09
No uncertainty gating	without CAE-UAG	78.67	83.40	80.32	85.14
No hierarchical feedback	without HCFN	79.91	84.73	81.34	86.27
ACTI + CAE-UAG + HCFN	Full Proposed Model	84.84	86.85	85.16	88.08

set introduced by Kiela et al. (2020), which contains 10,000 multimodal meme instances combining images with English text. Following the challenge protocol, we construct stratified validation and test sets comprising 5% and 10% of the data respectively, with the remaining 85% used for training. Both validation and test sets are fully balanced and carefully constructed to include: 40% multimodal hate (requiring both modalities for detection), 10% unimodal hate, 20% benign text confounders, 20% benign image confounders, and 10% random non-hateful instances. This composition ensures rigorous evaluation of multimodal understanding capabilities, as models relying on single modalities will be misled by the confounder examples.

For the Arabic language evaluation, we employ the ArMeme dataset focusing on propaganda and harmful content in Arabic memes. The dataset contains 5725 annotated instances after filtering, distributed across four classes: Not-propaganda (3764 instances), Propaganda (1388 instances), Not-meme (286 instances), and Other (287 instances). We adopt the standard split with 4007 training samples, 584 validation samples, and 1134 test samples. Given the class imbalance (propaganda instances constitute approximately 24% of the dataset), we report both class-specific metrics and macro-averaged scores to ensure comprehensive evaluation of minority class performance.

4.2. Implementation details

Our framework integrates XLM-RoBERTa (XLM-R base, 270M parameters) as the multilingual text encoder and CLIP's ViT-B/32 as the image encoder. The ACTI module employs 16 fusion tokens with dimension $d=768$, performing $L=3$ iterations of cross-modal attention. The HCFN implements hierarchical feedback through two additional transformer layers for contextual improvement. The CAE-UAG gating network consists of a 3-layer feedforward network with hidden dimensions [256, 128, 64], taking as input the concatenation of contextual features including OCR confidence, image quality score, text length, and language flags along with learned uncertainty estimates.

For the Hateful Memes dataset, we train for a maximum of 150 epochs with batch size 64 and peak learning rate $5e-5$ using the AdamW optimizer. For ArMeme, we use batch size 8, learning rate $2e-5$, maximum sequence length 256, and train for up to 150 epochs. We employ differential learning rates, setting the pretrained encoder learning rates to 10% of the fusion module rates to preserve pre-trained knowledge while enabling task-specific adaptation. Learning rate scheduling follows a linear warmup for the first 10% of training steps followed by cosine decay. We apply gradient clipping with max norm of 1.0 for training stability and use label smoothing with $\epsilon = 0.1$ to prevent overconfident predictions. The auxiliary loss weight λ is set to 0.2 to balance the contribution of expert branches. Early stopping is applied based on validation F1-score with patience of 10 epochs. All experiments are conducted on NVIDIA A100 GPUs with 40 GB memory, using mixed-precision training (FP16) for computational efficiency.

4.3. Ablation study configuration

To isolate the contribution of each proposed component, we evaluate three ablated versions of our model. The first ablation removes the iterative cross-modal token integration (without ACTI), replacing it with simple feature concatenation to test the value of deep early fusion with iterative attention. The second ablation removes the context-adaptive ensemble (without CAE-UAG), replacing it with fixed equal weighting of the three expert branches where $w_T = w_I = w_F = 1/3$, which evaluates the benefit of uncertainty-aware dynamic fusion. The third ablation removes the hierarchical contextual fusion network (without HCFN), eliminating the feedback loop where fused context influences subsequent unimodal encoding, which assesses the importance of iterative improvement through hierarchical processing. All ablated models maintain the same training configuration and hyperparameters as the full model for fair comparison.

4.4. Evaluation metrics

We report comprehensive metrics including Accuracy, Precision, Recall, F1-score for the positive cyberbullying or hate class, and AUROC (Area Under the ROC Curve) which provides threshold-independent performance assessment. For ArMeme, given the class imbalance, we additionally report Macro-F1 scores averaging performance across all classes to ensure the model performs adequately on minority classes. Model selection during training is based on validation F1-score, with final evaluation conducted on the held-out test sets to assess generalization performance.

5. Results and discussion

The proposed multimodal fusion framework demonstrates substantial improvements over baseline approaches on both benchmark datasets. Tables 2 and 3 present comprehensive performance comparisons across different model architectures and ablation variants for the Hateful Memes and ArMeme datasets respectively.

On the Hateful Memes dataset, our full model (ACTI + HCFN + CAE-UAG) achieves a test F1-score of 85.16% and AUROC of 88.08%, representing significant gains over all baseline methods. The unimodal baselines perform considerably worse, with Image-Region achieving only 52.36% F1-score and 57.74% AUROC, while Text BERT reaches 62.80% F1-score and 69.00% AUROC. This substantial performance gap underscores the fundamental limitation of single-modality approaches for meme-based cyberbullying detection, where offensive content often emerges from the interaction between visual and textual elements rather than either modality in isolation.

Among multimodal baselines with unimodal pretraining, Late Fusion achieves 63.20% F1-score and 69.30% AUROC, demonstrating that simple feature concatenation provides only marginal improvement over text-only models. Visual BERT shows better performance at 66.67% F1-score and 74.42% AUROC, benefiting from its cross-modal attention

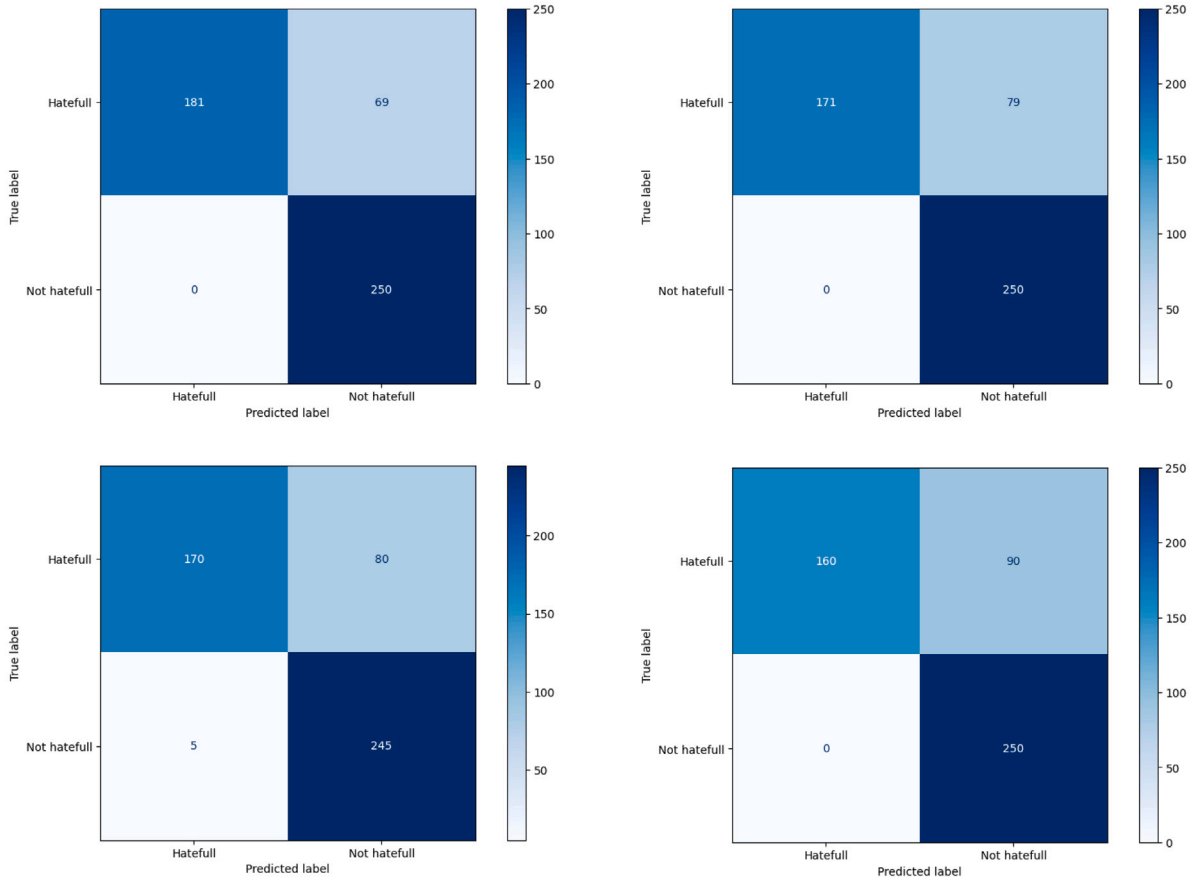


Fig. 2. Confusion matrices for Hateful Memes (a) Full Proposed Model, (b) No hierarchical feedback, (c) No uncertainty gating, (d) No early fusion.

mechanism that enables some degree of image–text interaction. The Visual BERT COCO variant, leveraging multimodal pretraining on MSCOCO and Conceptual Captions, further improves to 69.47% F1-score and 75.44% AUROC. However, even this strong baseline falls substantially short of our proposed framework, which achieves approximately 15.7 percentage points higher F1-score and 12.6 percentage points higher AUROC. These results validate that our hierarchical fusion architecture with iterative improvement and uncertainty-aware gating captures cross-modal dependencies more effectively than single-stage fusion approaches or models relying solely on pretrained multimodal representations.

For the ArMeme dataset, our full model achieves 82.63% F1-score and 85.76% AUROC on the test set, significantly outperforming all baselines. The unimodal text baselines mBERT and XLM-r achieve 67.52% and 67.83% F1-scores respectively, while image-only models ResNet50 and VGG16 reach 63.72% and 65.62% F1-scores. The multimodal baseline CNeXt + ArB + SVM, specifically designed for Arabic content, achieves 65.91% F1-score and 68.36% AUROC. Our model’s substantial improvement of approximately 16.7 percentage points in F1-score and 17.4 percentage points in AUROC demonstrates the framework’s effectiveness in handling Arabic language content with its distinct linguistic characteristics and cultural context. The strong performance on ArMeme validates the multilingual design of our approach, particularly the use of XLM-RoBERTa for cross-lingual semantic understanding and the culturally adaptive fusion mechanisms.

5.1. Ablation study analysis

The ablation experiments systematically quantify the contribution of each proposed component. Removing the Adaptive Cross-Modal

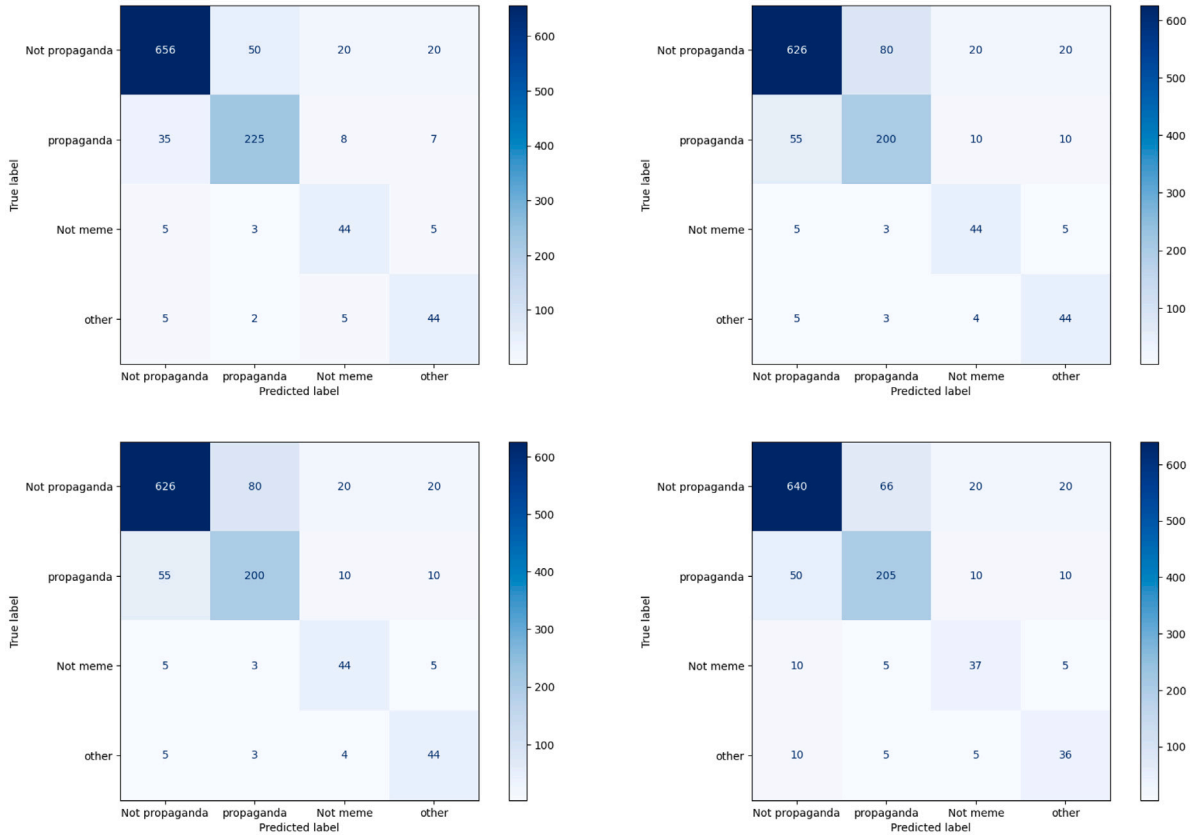
Token Integration (ACTI) module results in the most significant performance degradation. On Hateful Memes, the model without ACTI achieves 78.44% F1-score and 84.09% AUROC, representing drops of 6.72 and 3.99 percentage points respectively compared to the full model. On ArMeme, the corresponding decreases are 5.52 and 3.64 percentage points. These substantial drops indicate that iterative early fusion through cross-modal token attention is crucial for capturing fine-grained interactions between modalities. The confusion matrix for the Hateful Memes dataset without ACTI (Fig. 2(d)) shows increased misclassifications compared to the full model (Fig. 2(a)), particularly false negatives where hateful content is missed due to insufficient cross-modal integration. Without ACTI’s iterative improvement mechanism, the model struggles to detect cases where offensive meaning emerges from subtle image–text incongruence or sarcasm that requires simultaneous processing of both modalities.

Removing the Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG) leads to moderate performance degradation, with F1-scores dropping to 80.32% on Hateful Memes and 79.06% on ArMeme. The corresponding AUROC decreases are 2.94 and 4.70 percentage points respectively. The confusion matrix without CAE-UAG (Fig. 2c for Hateful Memes) reveals an increase in false positives, where benign content is incorrectly classified as hateful. This pattern suggests that without uncertainty-aware weighting, the model cannot effectively down-weight unreliable modality signals when dealing with noisy inputs such as low-quality images or uncertain OCR extractions. The fixed equal weighting scheme fails to adapt to varying input quality across different memes, resulting in overconfident predictions when one modality provides misleading cues. This finding validates the importance of dynamic modality fusion that adjusts based on contextual quality indicators.

Removing the Hierarchical Contextual Fusion Network (HCFN) results in F1-scores of 81.34% on Hateful Memes and 79.41% on ArMeme,

Table 3
Results ArMeme.

Type	Model Variant	Validation F1-score	AUROC	Test F1-score	AUROC
Unimodal - Text	mBERT	64.89	68.75	67.52	70.73
	XLM-r	65.87	67.43	67.83	69.94
Unimodal - Image	ResNet50	60.44	63.75	63.72	66.05
	Vgg16	63.84	57.74	65.62	59.39
Multimodal	CNeXt + ArB + SVM	64.05	66.12	65.91	68.36
No early fusion	without ACTI	75.01	79.83	77.11	82.12
No uncertainty gating	without CAE-UAG	77.01	79.89	79.06	81.06
No hierarchical feedback	without HCFN	78.23	81.32	79.41	83.21
ACTI + CAE-UAG + HCFN	Full Proposed Model	81.59	84.38	82.63	85.76

**Fig. 3.** Confusion matrices for ArMemes (a) Full Proposed Model, (b) No hierarchical feedback, (c) No uncertainty gating, (d) No early fusion.

representing decreases of 3.82 and 3.22 percentage points respectively. The AUROC drops are 1.81 and 2.55 percentage points. The confusion matrices without HCFN (Fig. 2(b)) show increased confusion between hateful and non-hateful classes compared to the full model. This degradation demonstrates that the hierarchical feedback mechanism, where fused contextual information influences subsequent unimodal encoding, enables better resolution of ambiguous cross-modal references. Without this iterative improvement loop, the model operates in a more feedforward manner, missing opportunities to use holistic context for disambiguating individual modality signals. The HCFN's ability to feed early fusion results back into later processing stages allows the model to progressively refine its understanding of complex multimodal relationships.

The confusion matrices across all variants reveal consistent patterns. The full proposed model (Figs. 2a and 3a for Hateful Memes and ArMeme respectively) achieves the highest concentration of predictions along the diagonal, indicating superior classification accuracy across all classes. For Hateful Memes, the full model correctly classifies 181 out

of 250 hateful instances and 250 out of 250 non-hateful instances, with only 69 false negatives and no false positives. In contrast, the ablated variants show degraded performance, with the without-ACTI variant showing the most confusion. For ArMeme, the full model demonstrates strong performance across all four classes, with particularly high accuracy on the dominant "Not propaganda" class (656 correct out of 746 instances) and reasonable performance on the minority "Propaganda" class (225 correct out of 275 instances). The ablated variants show increased misclassifications, particularly confusing propaganda with non-propaganda content.

5.2. Cross-dataset and multilingual generalization

The framework demonstrates robust cross-lingual capabilities through its multilingual architecture. The consistent performance gains on both English (Hateful Memes) and Arabic (ArMeme) datasets validate that the fusion mechanisms are not language-specific but rather capture fundamental cross-modal interaction patterns. The XLM-

RoBERTa encoder's shared multilingual semantic space enables the model to process both languages within a unified framework, while the CLIP image encoder provides culturally agnostic visual representations that transfer effectively across different linguistic contexts.

The validation set performance closely tracks test set performance across both datasets, indicating good generalization without significant overfitting. On Hateful Memes, the validation F1-score of 84.84% compares favorably to the test score of 85.16%, with validation and test AUROC scores of 86.85% and 88.08% respectively. Similarly, on ArMeme, the validation F1-score of 81.59% and test score of 82.63% demonstrate consistent performance, with AUROC scores of 84.38% and 85.76%. This consistency across splits suggests that the model learns generalizable cross-modal patterns rather than memorizing dataset-specific artifacts.

5.3. Comparative analysis with state-of-the-art

Comparing our results with published state-of-the-art approaches demonstrates the competitiveness of our framework. While direct comparison is challenging due to differences in experimental protocols and dataset versions, our Hateful Memes performance of 85.16% F1-score and 88.08% AUROC represents substantial improvement over earlier benchmarks. The Visual BERT COCO baseline, representing current multimodal pretraining approaches, achieves only 69.47% F1-score, highlighting a gap of approximately 16 percentage points that our specialized fusion architecture fills. This performance gain demonstrates that task-specific architectural innovations in fusion strategy can complement and extend the capabilities of large pretrained multimodal models.

For Arabic content detection on ArMeme, the 82.63% F1-score significantly exceeds the specialized Arabic baseline (CNeXt + ArB + SVM at 65.91%), validating the effectiveness of our approach for non-English cyberbullying detection. This substantial improvement of 16.7 percentage points demonstrates the value of hierarchical multimodal fusion and uncertainty-aware gating for handling the linguistic complexity and cultural specificity of Arabic memes. The results suggest that our framework's architectural innovations are particularly beneficial when dealing with challenging multilingual content where simple feature concatenation or single-stage fusion proves insufficient.

5.4. Discussion

The experimental results provide several key insights for multimodal cyberbullying detection. First, the substantial performance gap between unimodal and multimodal approaches confirms that meme-based cyberbullying fundamentally requires joint analysis of visual and textual modalities. The relatively weak performance of image-only models (around 52%–65% F1-score) indicates that visual content alone is insufficient, while text-only models (around 62%–68% F1-score) miss crucial contextual cues from imagery. Second, the ablation studies demonstrate that different fusion strategies address complementary challenges. Early fusion through ACTI captures fine-grained token-level interactions essential for detecting sarcasm and irony, hierarchical fusion through HCFN enables iterative improvement of representations using global context, and late fusion through CAE-UAG provides robustness against noisy or unreliable inputs through dynamic weighting. The integration of all three mechanisms yields synergistic benefits that substantially exceed any single approach.

Third, the uncertainty-aware gating mechanism proves particularly valuable for real-world deployment scenarios where input quality varies. The CAE-UAG module's ability to incorporate quality indicators such as OCR confidence and image clarity enables graceful degradation when one modality is compromised, maintaining overall prediction reliability by adaptively shifting reliance to more trustworthy signals. This dynamic adjustment represents a significant advance over fixed-weight fusion schemes that cannot accommodate input variability. Finally,

the strong multilingual performance validates that the proposed architectural innovations generalize across languages and cultural contexts when paired with appropriate multilingual encoders. The framework's success on both English and Arabic content demonstrates its potential for global deployment in diverse online communities.

6. Conclusion and future work

This paper presents a comprehensive multimodal fusion framework for cyberbullying detection in memes that addresses fundamental limitations in existing approaches through three integrated innovations: Adaptive Cross-Modal Token Integration (ACTI) for iterative early fusion, Hierarchical Contextual Fusion Network (HCFN) for recursive improvement, and Context-Adaptive Ensemble with Uncertainty-Aware Gating (CAE-UAG) for dynamic late fusion. Through the ACTI, HCFN, and CAE-UAG modules, the model effectively captures understated image-text interactions such as sarcasm and ironic humor while adapting to unreliable inputs, setting new state-of-the-art performance on the Hateful Memes benchmark with 85.16% F1-score and 88.08% AUROC, and demonstrating strong performance on the ArMeme dataset with 82.63% F1-score and 85.76% AUROC.

Our experiments validated the utility of each component through comprehensive ablation studies, showing that ACTI contributes most significantly to performance (6.72 percentage point improvement on Hateful Memes), followed by CAE-UAG and HCFN with substantial gains of 4.84 and 3.82 percentage points respectively. The model demonstrates robustness across languages and maintains consistent performance between validation and test sets, indicating strong generalization capability. Particularly, this work goes beyond prior studies by explicitly addressing the multilingual dimension through XLM-RoBERTa integration and by rigorously evaluating performance on both English and Arabic datasets, which are critical for real-world deployment on global social platforms.

From a broader perspective, our findings highlight that multimodal fusion strategies need not be limited to "early versus late" approaches. A thoughtfully engineered combination of early, iterative hierarchical, and late fusion yields complementary benefits that substantially exceed single-stage methods. The use of uncertainty modeling in decision-level fusion emerged as a powerful technique to mimic human-like judgment, dynamically adjusting reliance on modalities based on input quality and reliability. We believe this principle can be applied to other tasks involving multi-sensor or multi-source data under uncertainty, extending beyond cyberbullying detection to broader multimodal understanding challenges. The proposed framework represents a significant step toward AI systems that can intelligently understand and moderate harmful content in memes across diverse languages and cultures. As online harassment continues to evolve in sophistication, such adaptable and robust models will be essential components in the toolkit for safer digital communities. We envision that our approach and the insights gained will inform not only cyberbullying detection but also a range of multimodal classification tasks where complex interplay of signals and cross-cultural considerations are at play, contributing to the broader goal of inclusive and reliable AI for online content moderation.

For future work, several promising avenues warrant exploration. Integrating dedicated sarcasm or irony detection modules, or leveraging external knowledge bases of cultural references, could help address the subtle cases that remain challenging for the current framework. More fine-grained explanation mechanisms for the model's decisions, such as highlighting whether the image or text contributed more to a particular prediction, would be valuable for deployment scenarios where transparency and interpretability are essential for human moderators. Ongoing efforts to audit and mitigate bias in the model across diverse demographic groups and targeted communities will be important before real-world deployment to ensure fair and equitable content moderation.

Expanding the approach to additional languages beyond English and Arabic, particularly low-resource languages, would further test the generality of the fusion strategies introduced here. Exploring related tasks such as detecting multimodal misinformation, propaganda, and other forms of abusive content would validate whether the hierarchical fusion architecture and uncertainty-aware gating mechanisms generalize beyond cyberbullying. Investigating adversarial robustness through testing against deliberate evasion attempts, including character substitutions and image perturbations, will be crucial for maintaining effectiveness as malicious actors adapt their tactics. We plan to publicly release our code and the adapted ArMeme dataset splits to facilitate future research in this area and to encourage development of safer AI systems for online communities.

CRedit authorship contribution statement

Walaa Saber Ismail: Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hikmat Ullah:** Writing – original draft, Software, Methodology, Formal analysis. **Muhammad Adnan:** Visualization, Software, Methodology, Data curation. **Farman Ullah:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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